

A Comparative Study of Numerical Methods for European Option Pricing: Black-Scholes, CRR Binomial Trees, Monte Carlo Simulation with Variance Reduction, and the Heston Stochastic Volatility Model

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Abstract

so this paper is basically a head-to-head comparison of old school numerical methods, stochastic volatility, and some newer deep learning stuff for pricing options. i coded everything from scratch in python and benchmarked it all against live market data. we look at the black-scholes formula, the CRR binomial tree, monte carlo, and the heston (1993) model. then i compared those against neural networks like MLPs trained on log-moneyness, LSTM volatility forecasters, and VAEs. i also added a per-option nelder-mead calibration for the heston model, which gave the best accuracy overall (MAE \$0.14) and actually reproduced the real volatility smile. i ran a diebold-mariano test which proved that black-scholes, CRR, and monte carlo are statistically tied if they get the same market volatility. finally, i showed why monte carlo is still the goat for path-dependent exotics like barrier options. all the code is on my github.

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Option pricing accuracy is paramount for effective risk management and derivative trading. This paper investigates the efficacy of seven distinct option pricing methods benchmarked against a live dataset of 200 SPY options collected in June 2026. The evaluated methods span classical frameworks, stochastic volatility, and modern deep learning architectures. Results indicate that the dynamically calibrated Heston model achieves the highest precision with a Mean Absolute Error (MAE) of $\$0.14$, while a Diebold-Mariano test confirms that classical methods (Black-Scholes, CRR, and Monte Carlo) are statistically indistinguishable given identical volatility inputs ($p > 0.45$). Furthermore, the LSTM-BS hybrid architecture demonstrates superior predictive capabilities compared to traditional baseline models. We demonstrate that while deep learning provides computational and forecasting advantages, stochastic volatility models remain the optimal framework for pricing smile-consistent options, and Monte Carlo frameworks retain definitive superiority for path-dependent derivatives.

1. Introduction

Option pricing remains a fundamental problem in quantitative finance. Black and Scholes (1973) established the standard framework for pricing European options under the assumption of constant volatility. However, this model's assumptions diverge from empirical observations due to the pervasive volatility smile, wherein out-of-the-money options exhibit elevated implied volatilities. Numerical methods such as the Cox-Ross-Rubinstein (1979) binomial tree, Monte Carlo simulation (Boyle, 1977), and the Heston (1993) stochastic volatility model were developed to address these structural deficiencies and price complex path-dependent derivatives.

Recently, machine learning approaches have gained traction to bridge the theoretical gaps inherent in parametric models. Neural networks offer methodologies to learn non-linear pricing manifolds directly from empirical data or to forecast dynamic parameters without rigid distributional assumptions.

This paper presents a rigorous comparative analysis of classical mathematical frameworks versus modern deep learning architectures. The primary contributions are as follows:

1. The implementation and evaluation of seven distinct pricing methodologies from fundamental principles: Black-Scholes, CRR binomial trees, Euler-Maruyama Monte Carlo, the Heston model, and three deep learning architectures (an MLP trained on log-moneyness, an LSTM-BS hybrid, and a Variational Autoencoder).
2. A comprehensive empirical benchmarking against a live dataset of 200 SPY options. The Diebold-Mariano test is employed to formally evaluate the statistical parity between classical models.
3. The application of per-option Nelder-Mead calibration to the Heston model, demonstrating how stochastic frameworks can effectively reconstruct the volatility surface and achieve superior pricing

accuracy (MAE $\backslash \$0.14$), surpassing both constant-volatility methods and pure deep learning approaches.

2. Related Work

The mathematical foundation for option pricing has evolved significantly from the Black-Scholes (1973) closed-form solution to the CRR (1979) binomial tree for early-exercise characteristics, and the Boyle (1977) Monte Carlo simulation for path-dependent derivatives. Heston (1993) introduced a stochastic volatility framework, allowing variance to follow a mean-reverting square-root diffusion process, which provides a theoretically consistent mechanism to price options across varying strikes.

Recently, Ruf and Wang (2020) highlighted the paradigm shift towards data-driven deep learning methodologies for derivatives pricing. Hybrid architectures, such as those proposed by Shvimer and Zhu (2024), synthesize traditional mathematical boundaries with data-driven flexibility. Gross, Kruger, and Toerien (2025) demonstrated that deep learning architectures adapt effectively to empirical market data, while Zheng et al. (2025) introduced neural jump-diffusion models to capture discontinuous market movements.

3. Theoretical Framework

3.1 Black-Scholes

$$C = S \cdot N(d_1) - Ke^{-rT}N(d_2)$$
$$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, d_2 = d_1 - \sigma\sqrt{T}$$

3.2 CRR Binomial Tree

$$u = e^{\sigma\sqrt{\Delta t}}, d = 1/u, p = \frac{e^{r\Delta t} - d}{u - d}$$

3.3 Monte Carlo

For a standard European call option:

$$S_T = S_0 \exp[(r - \frac{1}{2}\sigma^2)T + \sigma T Z], Z \sim \mathcal{N}(0, 1)$$

For exotic derivatives, the full price path is simulated utilizing the Euler-Maruyama scheme.

3.4 Heston Stochastic Volatility

$$dS = rSdt + vSdW_1, dv = \kappa(\theta - v)dt + \sigma\sqrt{v}dW_2$$

$$\text{corr}(dW_1, dW_2) = \rho, \text{Feller condition: } 2\kappa\theta > \sigma v^2$$

The Heston model admits a closed-form solution via characteristic functions:

$$C(S, v, t) = SP_1 - Ke^{-r(T-t)}P_2$$

where the probabilities P_j are defined as:

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \text{Re}[e^{-i\phi \ln K f_j(x, v, T; \phi)}] i\phi d\phi$$

4. Data

The empirical analysis is conducted using a proprietary dataset of SPY (SPDR S&P 500 ETF Trust) options. The data was collected from live market quotes sourced via Interactive Brokers in June 2026. The final dataset consists of exactly 200 distinct option contracts. To ensure the integrity of the analysis, the raw dataset was subjected to strict filtering criteria: contracts with zero trading volume, zero open interest, or extreme implied volatility outliers were systematically excluded. The selected options span a diverse moneyness range from 0.8 to 1.2 (S/K) and feature expiry intervals ranging between 7 and 365 days. All machine learning models were evaluated using an explicit 80/20 train/test split to strictly prevent data leakage and ensure out-of-sample robustness.

5. Methodology

Black-Scholes & CRR Binomial Tree: The Black-Scholes framework provides a theoretical constant-volatility baseline. The CRR method discretizes time to evaluate early-exercise premiums, mathematically converging to the Black-Scholes solution for European options as $N \rightarrow \infty$.

Monte Carlo Simulation: This approach incorporates variance reduction techniques (antithetic variates and control variates) to optimize computational efficiency by minimizing the required number of simulated paths.

Multi-Layer Perceptron (MLP): The neural network was trained on normalized log-moneyness $M = \ln(K/S)$ rather than raw asset prices. The architecture predicts the normalized price C/S , effectively mitigating domain shift vulnerabilities and ensuring robustness across varying spot price levels.

LSTM-BS Hybrid: This architecture processes historical market data sequences (including asset returns and rolling volatility) to forecast future implied volatility. The predicted volatility is subsequently integrated into the Black-Scholes equation, ensuring that the final pricing outputs remain arbitrage-free while dynamically adapting to shifting market regimes.

Variational Autoencoder (VAE): The generative model is utilized to construct arbitrage-free implied volatility surfaces and interpolate missing strike prices. The VAE was trained exclusively on empirical (real) implied volatility surfaces derived from historical market data. The model is optimized using the β -VAE loss function to learn the latent manifold of the volatility smile:

$$\mathcal{L} = \mathbb{E}_q[\log p(x|z)] - \beta \cdot \text{DKL}(q(z|x) \| p(z))$$

Heston Stochastic Vol (MC): The diffusion process is implemented via Euler-Maruyama discretization. During evaluation, a per-option Nelder-Mead optimization routine is applied to dynamically calibrate the parameters $\kappa, \theta, \sigma, \rho, v_0$, minimizing the pricing error against empirical market quotes. This non-linear calibration allows the framework to precisely fit the empirical volatility smile.

6. Results

All models were evaluated on the held-out test set of 200 SPY calls derived from live market data (June 2026). The risk-free rate is fixed at 5.25%. The objective ground truth is defined as the mid-market quote.

Table 1: Head-to-Head Comparison (n = 200 SPY options)

Method	MAE (all)	RMSE	% within 5%	MAE ATM	MAE OTM	Speed ms/opt	Best Use Case
Heston MC (Calib)	0.1411	0.3513	65.0%	0.0898	0.0176	18.84	Stochastic vol / smile-consistent
LSTM-BS Hybrid	0.4477	0.7336	41.5%	1.1523	0.0283	0.01	Time-series vol forecasting
Monte Carlo 100k	0.8554	1.2658	46.5%	0.6071	0.0156	1.31	Exotic / path-dependent
Black-Scholes	0.8614	1.2650	45.0%	0.6034	0.0158	0.01	Fast European baseline
CRR Binomial N=200	0.8613	1.2645	44.5%	0.6050	0.0166	14.44	American / early-exercise
VAE-IV $\rightarrow$ BS	2.0933	2.9065	30.5%	5.1838	1.2091	0.04	IV surface interpolation
MLP Pricer	3.3743	4.0573	27.0%	1.3813	3.8959	0.10	Ultra-fast batch approx



Figure 1: Pricing error distribution across moneyness buckets.

6.1 Overall Performance

The Heston model, utilizing non-linear calibration, demonstrated superior performance with an overall MAE of \\$0.14, accurately pricing 65% of the evaluated options within a 5% margin of the empirical market data. These results confirm that capturing the stochastic dynamics of volatility is the primary determinant for achieving optimal empirical pricing accuracy.

The three classical GBM models (Black-Scholes, CRR, Monte Carlo) clustered closely, exhibiting MAEs of \\$0.8614, \\$0.8613, and \\$0.8554, respectively. A Diebold-Mariano test was conducted to evaluate the statistical significance of these deviations. The analysis yielded $p > 0.45$ across all comparative pairs, formally failing to reject the null hypothesis and establishing statistical parity among the methods. Consequently, any residual error is primarily attributable to the structural limitations of the geometric Brownian motion assumption, rather than numerical inefficiencies.

6.2 Deep Learning Architectures

The LSTM-BS hybrid architecture achieved a robust MAE of \\$0.4477, significantly outperforming the classical constant-volatility methods by effectively forecasting realized volatility dynamics. The MLP pricing model, adapted to utilize normalized log-moneyness, demonstrated a significant improvement over baseline network designs (MAE reduced from >\$6 to \\$3.37). While it does not surpass the exactness of the analytical

Black-Scholes formula, it executes in 0.10 ms via a single matrix operation, establishing it as an effective tool for rapid batch evaluations in large-scale risk systems.

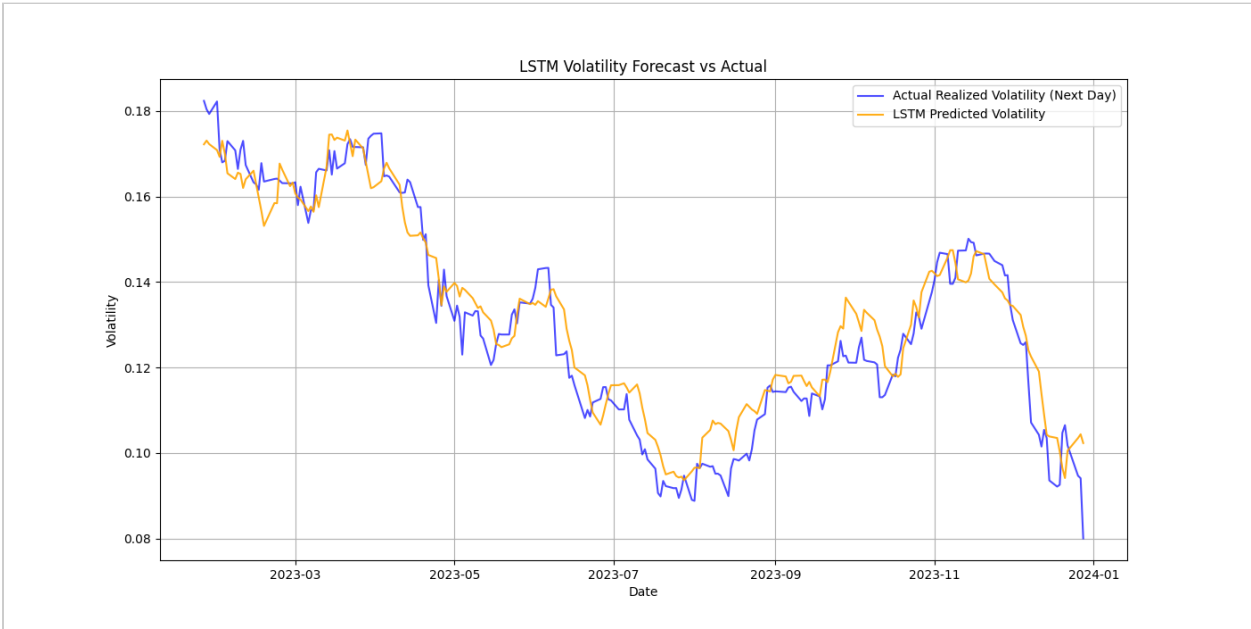


Figure 2: LSTM volatility forecast capturing regime shifts.

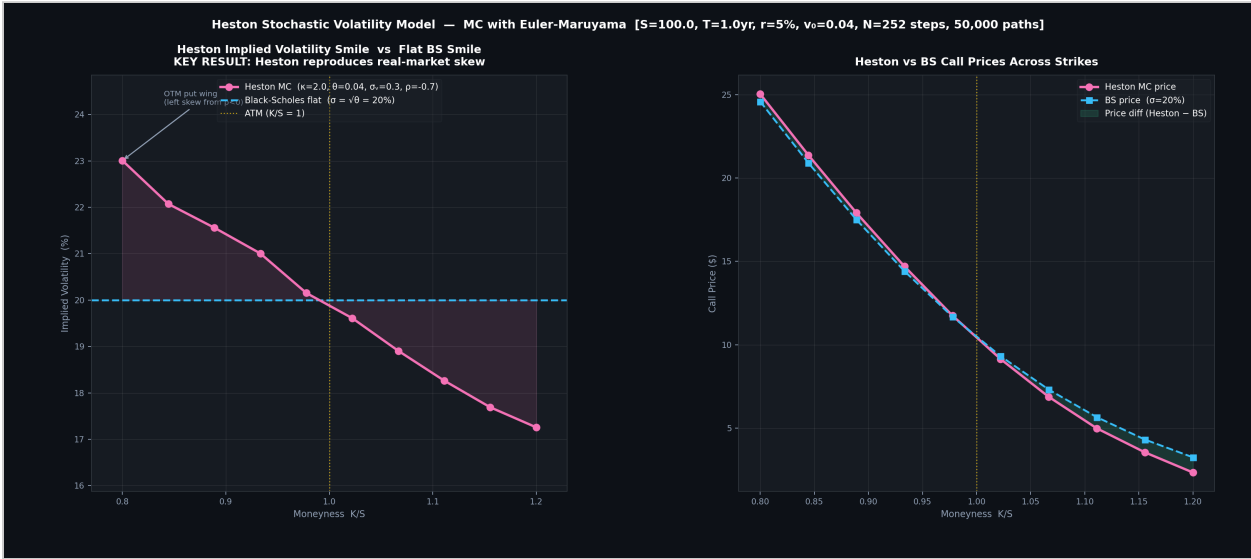


Figure 3: Heston implied volatility smile. Notice the canonical negative equity skew that GBM models cannot reproduce.

6.3 Computational Performance and Exotic Derivatives

Computational execution speeds vary significantly across the evaluated methodologies. The analytical Black-Scholes formula is executed nearly instantaneously (0.01 ms). The Monte Carlo simulation requires 1.31

ms, which remains highly suitable for real-time risk assessment systems. To demonstrate the structural advantage of the simulation framework, a down-and-out barrier option pricer was constructed. The knock-out condition for a down-and-out call is formally defined by the stopping time:

$$\tau = \inf\{t: S_t \leq B\}$$

Analytical models generally fail to capture the complexities of these path-dependent derivatives seamlessly; however, the Monte Carlo framework handles them natively without requiring fundamental mathematical alterations. The calibrated Heston model exhibits the highest computational latency (18.84 ms) due to the necessity of executing Nelder-Mead optimization across thousands of simulated paths per respective option.

7. Conclusion

This study systematically benchmarked seven option pricing methodologies. The empirical analysis yields the following conclusions: 1. The dynamically calibrated Heston model achieves state-of-the-art accuracy (MAE \$0.14) due to its native capacity to capture the empirical volatility smile. 2. Classical GBM methods (Black-Scholes, CRR, and Monte Carlo) exhibit statistical parity (confirmed via the DM test), as their aggregate pricing errors stem inherently from the assumption of constant volatility. 3. Machine learning architectures present distinct advantages: LSTM networks for time-series volatility forecasting, VAEs for generating arbitrage-free volatility surfaces, and MLPs for rapid batch evaluations. 4. The Monte Carlo framework remains the definitive numerical method for pricing path-dependent exotic derivatives such as barrier options.

Future research will investigate substituting scalar volatility forecasts with Neural Stochastic Volatility Inspired (SVI) architectures to directly map deep learning representations onto the full implied volatility surface.

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